Commission to consider the potential increased message traffic that could result from the proposed minimum pricing increments and stated that the proposal would result in a significant increase in message traffic.[237] The commenter recommended the Commission take a measured and phased approach for reducing the minimum pricing increment for quoting to apply the minimum quoting increment initially to a limited number of stocks and additional groups of stocks in subsequent phases, with review of market resiliency during each phase.

The amendments modifying Rule 612 will result in less message traffic, fewer systems changes and lower costs related to updating ticks for NMS stocks compared to the original proposal and therefore there should pose less of a concern related to market resiliency. The modified amendment adopts a single sub-penny increment that impacts a smaller universe of NMS stocks compared to the proposal, which included three sub-penny increments that would have impacted more NMS stocks. The need for a phased approach is significantly reduced because fewer NMS stocks will be impacted by the one additional minimum quoting increment, and there will be fewer ticks between the spread.

The commenter stated that the potential costs to industry members from increased message traffic would include purchasing additional computer hardware such as servers and that the costs would also apply to production, backup, test, and development environments.[238] The commenter stated that the actual costs would be multiples of the estimated costs from the proposal. However, the adopted amendment to Rule 612 will result in less message traffic than the proposal because it has fewer quoting increments. Consequently, the modified amendments

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237 See FIF Letter at 7. See also Robinhood Letter at 41; Morgan Stanley Letter at 3; UBS Letter at 12; Citigroup Letter at 4; TradeStation Letter at 7; and Goldman Sachs Letter at 9.

238 See FIF Letter at 9. See also Citigroup Letter at 2. See infra section VII.D.5.a.

that are being adopted will reduce computer hardware and developmental costs for the industry compared to the proposal. In the Proposing Release, the Commission considered the message traffic of the options markets, and the systems for the options markets that handle many times more messages compared to (1) the current NMS stock market or (2) the estimated additional message traffic from the adopted amendments.[239] One commenter submitted data that supported this conclusion.[240]

The commenter also raised concerns that increased quote message traffic could significantly increase the costs of the operation of the CAT system.[241] The commenter recommended that the Commission estimate the potential increase in message traffic, provide those estimates to CAT LLC, obtain estimates from the CAT LLC of the increased CAT costs that would result from this increased message traffic, and factor the estimated costs into the cost benefit analysis of the proposed minimum pricing increments changes. Another commenter also stated that the Commission failed to consider whether the increase in message traffic will increase the CAT operating budget.[242] The Commission estimates the impact of the adopted amendments on message traffic, and thus on the CAT operating budget in section VII.D.1.c. As

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239 See Proposing Release, supra note 11, at 80279, notes 196 and 197 (stating that in the second quarter of 2011, the average peak message per second for Tapes A and B reported by the CTA/CQ Plan was 1,015,000 and for Tape C reported by the UTP Plan was 408,300 versus 36.4 million reported by the Options Price Reporting Authority ("OPRA")). See also section VII.E.1.

240 See NYSE Letter I at 11-13.

241 See FIF Letter at 10 ("FIF members are concerned that increased message traffic could significantly increase the costs for the operation of the CAT system as increased quote volumes (including increased frequency of quote updates) would increase the number of CAT-reportable events. 100% of these increased CAT costs would be charged to broker-dealers and exchanges. The operating expenses for CAT were $84.5 million for 2020 and $146.5 million for 2021. CAT LLC, the operator of the CAT system, has estimated the total expenditures for CAT for 2022 at $178.9 million. These costs are in excess of the costs that were contemplated in the CAT NMS Plan.").

242 See Citadel Letter II at 5. The commenter added that increased message traffic increases costs for all market participants, including higher fees charged by CAT and the exclusive SIPs. See also Citadel Letter I at 9 and Virtu Letter II at 6-7.

discussed further below, the Commission estimates the increase in CAT costs associated with adopting the additional minimum pricing increment to be approximately $4.1 million per year.[243] The Commission does not believe it is appropriate to delay action on Rule 612 to have CAT LLC engage in its own analysis of the potential costs.

Commenters raised the issue of increased market data volume on competing consolidators, which are not yet in operation.[244] Likewise, the possible costs to potential competing consolidators will be reduced vis-à-vis the proposal. The Commission recognizes that while the costs may be lower than the proposed rule, the adopted rule could nevertheless create increased message traffic than the preexisting rule. It follows that more message traffic could lead to more possible costs for competing consolidators. However, this new message traffic should still be within the operational capacity of the existing computer systems.[245]

One commenter stated that even with the largest potential increases in messages, equity messaging traffic would remain well below that of the options market and that "the increase in messaging activity from adopting finer tick increments is now well within the industry's capability."[246] On the other hand, another commenter stated that a larger number of ticks across a large number of stocks would lead to increased message traffic, which would, in turn, increase data and infrastructure costs and market latency.[247] One commenter added that increased message traffic would lead to increased latency, which would harm market participants by

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243 See infra section VII.D.1.c.

244 See, e.g., Citadel Letter II at 9 ("A material increase in total message traffic increases costs for all market participants, including due to the resulting higher fees charged by industry utilities, such as the [CAT] and the [SIP]") and Virtu Letter II at 6-7 ("The Commission has failed to analyze the impact of the significantly increased volume of market data on competing consolidators.").

245 See infra section VII.D.1.

246 See NYSE Letter II at 11 (stating that OPRA handles many times more messages than the equity markets).

247 See MFA Letter at 11.

disrupting trading strategies and impairing market functionality and liquidity.[248] As stated above, the adopted amendment to Rule 612 is significantly less complex than the proposal and will not result in the larger number of ticks across a large number of stocks as the commenter suggested. The proposal's four minimum tick increment has been simplified to one additional new tick at $0.005, and the proposal's reduction of minimum pricing increments for NMS stocks that had a TWAQS of $0.04 or less has been reduced to those NMS stocks that have a TWAQS equal to or less than $0.015, which results in fewer expected NMS stocks being assigned a smaller minimum pricing increment.[249] These adopted changes may result in significantly less message traffic than under the commenter's assumption on the proposal. While message traffic may increase over today's message traffic, any increase in message traffic will be significantly less than in the options market, and the options market participants have over the years adjusted to increasingly higher message traffic.[250]

6. Comments on Proposed Criteria for Assigning Minimum Pricing Increments

The Commission proposed to measure the TWAQS when determining the appropriate minimum pricing increment for NMS stocks and proposed four ranges of the TWAQS to determine the corresponding minimum pricing increments. The four proposed TWAQS ranges

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248 See Tradeweb Letter at 2-3 ("Even trading platforms with the most advanced technological infrastructure will need to expend considerable amounts of time and resources to prepare the accommodate increased message traffic, since any increase in latency (even at the millisecond level) would disrupt trading strategies, impair market functionality and liquidity, and, ultimately, harm market participants."); see also Virtu Letter II at 6 ("This increase in message traffic…will significantly add to the overall content of market data."). See also NYSE Letter I at 6 and Nasdaq Letter I at 9 ("Securities with too many ticks not only have wider spreads, but they also have more odd lots, and more message traffic, leading to a more fragile NBBO.").

249 See infra section VII.D.1.a.

250 See Options Clearing Corporation Daily Volume report, available at https://www.theocc.com/Market-Data/Market-Data-Reports/Volume-and-Open-Interest/Daily-Volume.

were: (1) equal to or less than \$0.008; (2) greater than \$0.008 but less than or equal to \$0.016; (3) greater than \$0.016 but less than or equal to \$0.04; and (4) greater than \$0.04. Preliminarily, the Commission believed that NMS stocks with a TWAQS of \$0.04 or less would have benefited from smaller minimum pricing increments. After considering the comments, the Commission is retaining TWAQS as the measure to determine when an NMS stock will be assigned smaller minimum pricing increment but has modified the threshold to be equal to or less than \$0.015.

Many commenters stated that tick-constrained stocks would benefit from smaller minimum pricing increments.[251] Commenters, however, raised concerns about reducing the minimum pricing increment for NMS stocks that were not experiencing tick constraint with the \$0.01 minimum pricing increment.[252] One commenter stated that "a reduction in tick sizes for those stocks that are merely near-tick-constrained will not result in meaningful price-improvements and will not be worth the increased risk of diminished liquidity to bids and offers being spread too thinly across too many price points."[253] As adopted, the new \$0.005 minimum pricing increment will be assigned to those NMS stocks that have a TWAQS of \$0.015 or less. These NMS stocks are experiencing constraint with the \$0.01 minimum pricing increment and

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251 See, e.g., Pragma Letter at 6 ("tick-constrained stocks will benefit from smaller tick sizes with narrower spreads."); MEMX Letter; NYSE, Schwab, and Citadel Letter; IEX Letter I at 7; Nasdaq Letter I at 2 ("Nasdaq supports adjusting the minimum pricing increment ("tick size") to better reflect the trading dynamics of Regulation National Market System ("Reg. NMS") securities."); Brandes Letter at 2; Schwab Letter II at 35; and Robinhood Letter at 46.

252 See, e.g., IEX Letter I; Pragma Letter; Invesco Letter; ICI Letter II at 14 (stating that the Commission should not apply sub-penny increments to stocks that are not tick-constrained); ASA Letter ("we strongly oppose the application of a one-half cent tick size to any stock outside of the most liquid (narrower spread) stocks."); Nasdaq Letter I at 14 ("We propose that securities fall into this new \$0.005 tick bucket only if they are tick-constrained."); and Cboe Letter II.

253 See Invesco Letter at 3. See also e.g., ICI Letter II (stating that there is no market failure or harm identified for stocks that are not tick-constrained.) and Brandes Letter at 2 (favoring a reduction to \$0.005 for those stocks that are experiencing constraint with the \$0.01 increment and stating that the proposed reduction in a minimum pricing increment for stocks that had a TWAQS of \$0.04 or less was too broad).

will benefit from being able to be quoted in the smaller increment. As adopted, the Commission has modified the amendment so as not to assign the smaller $0.005 increment to those NMS stocks that are not necessarily experiencing constraint with the $0.01 minimum pricing increment.

Some commenters stated that the TWAQS of $0.011 should be used for identifying NMS stocks that are experiencing tick constraint.[254] However, one commenter recommended that NMS stocks that "could easily become tick-constrained" should have their minimum pricing increment reduced.[255] Other commenters offered other recommendations as to the TWAQS threshold for reducing minimum pricing increments, including a TWAQS threshold of $0.02 or less,[256] a TWAQS threshold of $0.016 or less,[257] and a TWAQS threshold of 0.015 or less.[258]

As discussed further below, the Commission is adopting the TWAQS threshold of $0.015 or less in order to identify NMS stocks that will be eligible for the $0.005 minimum pricing increment.[259] This amendment will generally result in these NMS stocks having a bid-ask spread

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254 See, e.g., NYSE Letter I; Vanguard Letter; Cboe Letter I; and Schwab Letter II at 35-36. But see also Invesco Letter at 3 (stating that $0.011 was overly broad and would result in unnecessary tick reductions for stocks that are not tick-constrained.).

255 See, e.g., IEX Letter I at 7 ("IEX agrees with the premise that tick sizes should be reduced for stocks that are currently "tick-constrained" or could easily become tick-constrained because of the current one-cent limitation.").

256 See IEX Letter I at 7, 13 ("We believe that reducing the tick size and applying it to all securities with a TWAQS up to two cents will substantially improve the efficiency of displayed trading . . .").

257 See BMO Capital Letter at 2 and Form Letter Type H, of which 853 comments were received, available at https://www.sec.gov/comments/s7-30-22/s73022.htm.

258 See Pragma Letter at 6 ("While perhaps not conclusive, the lines of evidence from our analysis also suggest that the Proposal's range of 4 to 8 ticks is too many and will force wider spreads and higher trading costs on the market than necessary. This leads to our primary recommendation: stocks should be moved to a smaller tick size only when their average spread is less than 1/5 in the preceding month; and moved to a larger tick size only when their spread is greater than 4 ticks in the preceding month.").

259 See infra section VII.D.1.b for more discussion on TWAQS.

with one to three ticks, which will improve market quality.[260] Data analysis supports that liquidity and market quality will improve if NMS stocks with a TWAQS of $0.015 or less are assigned to the $0.005 minimum pricing increment.[261] Commenters provided analysis and cited studies that suggest that 2 to 4 ticks intra-spread is optimal for trading, which is consistent with the results of the Commission's analysis.[262]

Some commenters agreed that the TWAQS was the appropriate measure for determining the relevant minimum pricing increment.[263] Several commenters stated that as many stocks as possible should be identified as eligible for a smaller tick size.[264] Other commenters suggested that a multi-factor approach be taken in evaluating whether to reduce the minimum pricing increment for certain NMS stocks.[265] Commenters suggested that such factors include average

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260 See infra note 1303 and accompanying text. See also Nasdaq Letter I at 18 (stating that "quoting outside of the optimal 2-3 tick spreads leads to queues for tick-constrained securities and slower price formation for securities with overly-wide spreads.").

261 See infra section VII.D.1.b.

262 See, e.g., Nasdaq Letter I at 8, 18; Pragma Letter at 1; RBC Letter at 3; CCMR Letter at 23; Letter from Eric Swanson, Chief Executive Officer, XTX Markets LLC, dated Mar. 30, 2023 ("XTX Letter") at 4; MMI Letter at 5; and Harris Letter at 7. See also infra notes 1293-1299 and accompanying text.

263 See, e.g., IEX Letter I at 7 ("[w]e agree that TWAQS is a reasonable and appropriate measure to define which securities should be subject to a narrower tick size."); MEMX Letter; and BMO Capital Letter.

264 See, e.g., Form Letter Type K, of which 22 comments were received, available at https://www.sec.gov/comments/s7-30-22/s73022.htm; Anonymous Letter dated Mar. 6, 2023; letter from Victor Piousbox dated Mar. 6, 2023; letter from Jimit Raithatha dated Mar. 7, 2023; letter from Munib Mian dated Mar. 7, 2023; letter from Peter Unum dated Mar. 19, 2023; letter from Anonymous dated Mar. 22, 2023; and letters from Chris and Donna Graves, dated Mar. 26, 2023; Spencer Neukam dated Mar. 26, 2023; Samuel Cressy dated Mar. 24, 2023; and Zaf Khan dated Mar. 24, 2023.

265 See, e.g., Cboe Letter II at 3 ("The most critical step in any tick-size regime reform is first establishing an objective methodology designed to address truly tick-constrained securities. In this regard, we recommend using a multi-factor methodology, such as Cboe's Tick Size Reduction Framework."); Themis Letter at 7 (supporting Cboe's methodology); Cboe, State Street, et al. Letter; Optiver Letter (discussing the European Union's tick regime as considering stock price and liquidity); NYSE Letter I; ICI Letter II at 11 (stating that applying other factors would lessen concerns about an overbroad tick reduction and mitigate concerns about an adverse market outcome); BlackRock Letter at 5 (stating that quoted spread is one-dimensional and does not provide sufficient context for determining the optimal tick size); Citigroup Letter (recommending a new $0.005 quoting increment for the most liquid tick-constrained stocks); T. Rowe Price Letter (stating that a multi-factor approach would allow the Commission to measure whether a tick size is properly calibrated); STA Letter at 6 (recommending that the Commission use a multifactor

quoted size,[266] ratio of average quoted size to average traded size,[267] daily traded volume,[268] queue length,[269] quotes on multiple exchanges,[270] or stock price.[271] One commenter recommended the inclusion of factors such as large quoted displayed size and a relatively high level of liquidity based on average daily trading volume.[272]

One commenter suggested that in addition to the TWAQS, "quote stability" should be measured.[273] According to the commenter, quote stability would be measured by looking at a change in a stock's quote after execution; if the quote widens after an execution, "the quoted liquidity may not be sufficient for the liquidity demanded, suggesting that the quote increment is

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approach); Virtu Letter II at 6 (stating "one must consider many factors, not just quoted spread" and describing methods proposed by Cboe and Nasdaq); NYSE, Schwab, and Citadel Letter at 1 ("We define 'tick-constrained' to mean symbols that have an average quoted spread of 1.1 cents or less and a reasonable amount of available liquidity at the NBBO."); and Cambridge Letter at 6 (stating that securities should have an average quoted spread of 1.1 cents and be "reasonably liquid"). See also SIFMA Letter II at 36 ("SIFMA believes that a more robust analysis is necessary to evaluate the most appropriate tick sizes for purposes of achieving the best balance between available liquidity at the inside quotation versus narrower spreads.").

266 See, e.g., BlackRock Letter at 6 (". . . if material size was present at the National best Bid and Offer ('NBBO') or a significant proportion of executions were occurring at sub-penny prices, this would be a clear indication of fierce order book competition and interest to tighten the spread and trade in smaller increments."); T. Rowe Price Letter; and Citadel Letter I.

267 See, e.g., Cboe Letter II at 3 ("we started with the complete universe of NMS securities, and applied three constraints—quoted spread, quoted-size-to-trade-size ratio, and notional turnover ratio—to arrive at a group of securities that are quantifiably tick-constrained.") and BlackRock Letter.

268 See, e.g., BlackRock Letter at 5 ("BlackRock recommends that in addition to the time weighted quoted spread, the Commission should incorporate other factors for designating tick sizes, such as the average quoted size, ratio of average quoted size to average traded size, daily traded volume, or stock price."); Optiver Letter; T. Rowe Price Letter; and Citadel Letter I.

269 See, e.g., T. Rowe Price Letter at 4 ("Other factors that could be considered include queue length and quoted size at the top of the order book, turnover, and whether the stock is quoted on multiple exchanges.") and Citadel Letter I.

270 See, e.g., T. Rowe Price Letter at 4.

271 See, e.g., BlackRock Letter at 5 and Optiver Letter ("[w]e recommend that the Commission undertake further analysis of the optimal level of tick granularity, leveraging price and volume to define appropriate tick sizes.").

272 See ICI Letter I at 11. See also Cambridge Letter at (stating that minimum pricing increments should be reduced for those stocks that have a TWAQS of $0.011 or less and "are reasonably liquid.") and Citigroup Letter at 4.

273 See, e.g., NYSE Letter I at 3. See also B. Riley Letter.

not actually constraining quoting activity."[274] Another commenter suggested that the Commission consider using "spread leeway," which the commenter defined as equal to the average quoted spread divided by the minimum tick size.[275] The commenter stated that spread leeway could "effectively quantify the extent to which bid-ask spreads are constrained by the minimum tick size."[276] Another commenter suggested that in addition to a TWAQS of $0.011, there should be "balance or near equilibrium of multiple bids and offers at the top of the central order book" as this would "imply that market forces of supply and demand would naturally force the bid/ask spread tighter through market competition."[277] One commenter that recommended a multi-factor approach to identify NMS stocks suggested that in addition to average quoted spread, a high quoted size to traded size ratio and a high average daily notional turnover should be examined when identifying NMS stocks that are tick-constrained.[278] According to the commenter, a high quoted size to traded size ratio is "an objective signal that shows even though there is an abundance of liquidity, the current $0.01 tick constraint disincentivizes investors to cross the spread due to high costs, resulting in a lack of trade executions."[279] Further, the commenter stated that a high average daily notional turnover is "an objective signal because it

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274 Id.

275 See MMI Letter (stating that spread leeway… "quantifies the extent to which bid-ask spreads are constrained by the minimum tick size…is equal to the average quoted spread divided by the minimum tick size. Prior studies have suggested a spread leeway of 3-9 as optimal for tick sizes to be neither too small, nor too large.").

276 Id. at 5.

277 See Invesco Letter at 3.

278 See Cboe Letter I and Cboe Letter II at 3. See also Cboe, State Street, et al. Letter; State Street Letter at 3.

279 See Cboe Letter I at 2.

focuses the tick reduction effort on high turnover securities that would benefit from the ability to be traded in finer increments."[280] Other commenters supported this approach.[281]

After analyzing data to determine whether the suggested additional factors would be helpful in eliminating NMS stocks that could be harmed by a smaller minimum pricing increment,[282] the Commission has concluded that TWAQS is the appropriate measure to determine whether an NMS stock should be eligible for a smaller minimum pricing increment. Specifically, TWAQS provides a transparent and objective basis to determine whether the $0.01 minimum pricing increment results in a quoted spread that is too wide for a particular NMS stock. Other possible factors, such as average quoted size, ratio of average quoted size to average trade size, the average daily traded volume, queue length, quotes on multiple exchanges or stock price, would add unwarranted and additional complexity that would be difficult and costly for market participants to monitor because some of these measures require the purchase of proprietary data. Supplementing TWAQS with quoted size, turnover calculations, quote stability, and the other recommend criteria would similarly add additional complexity and responsibilities to the primary listing exchanges assigned to calculating TWAQS.[283] The additional criteria suggested by commenters are unnecessary because TWAQS is a sufficient, comprehensive and objective way to determine whether NMS stocks are experiencing issues of constraint related to the $0.01 minimum pricing increment for quotes and orders.[284] Specifically, the Commission concluded that harm is unlikely to result if the other data factors suggested by commenters (e.g.,

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[280] Id.

[281] See, e.g., SIFMA Letter II at 40; Tastytrade Letter at 18; and Themis Letter at 4.

[282] See infra section VII.D.1.b.iii. for further discussions of alternative criteria.

[283] See Proposing Release, supra note 11, at 80274.

[284] See infra section VII.D.1.b.iii.